Maxim Van de Wynckel

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I am a tech lead, researcher and project manager with a passion for interoperable (information) systems. My main expertise lies in linked open data, data spaces, sensor processing & RTLS. With my business skills and knowledge, I am capable of integrating innovation in existing business processes.

Employment

Vlaanderen connect.
2025 - now

Technical Team Lead

Via Vlaanderen connect. I work in the Department of Environment and Spatial Development primarily focussing on designing the data architecture and implementation for the industrial emissions portal regulation.

MVdW-Software
2019 - now

Software engineering consultant

Freelance software development and consultancy. Projects spanning from web development to IoT and AI integration into existing business processes. Project lead with external partners.

Vrije Universiteit Brussel

External Postdoctoral Researcher — 2025 - now

Unpaid external postdoctoral researcher at the Web & Information Systems Engineering Lab, focusing on interoperability, linked data, and dataspaces.

Academic and Teaching Staff — 2019 - 2025

Teaching assistant for various Bachelor and Master courses such as: Databases, Open Information Systems, Web Technologies, Information Visualisation, Next Generation User Interfaces, Advanced Topics in Big Data. Researcher at the Web & Information Systems Engineering Lab on interoperability, linked data, internet of things, augmented reality, human-computer interaction and applied artificial intelligence.

ChemStream BV.
2014 - 2018

Software engineer (student job)

Development of chemical database software.

UZ Brussel
2016

Software developer internship (student job)

Development of QR-code based appointment system. Optimising the printing of appointments via Zebra printers. Google Fit integration for patients. Healthcare software development.

BITPOWER BV.

Software developer internship (student job) — 2015

eID cloud authentication thesis preparatory training. Security testing and protocol analysis.

Education

Doctor of Sciences (PhD), Computer Science
2019 - 2025

Vrije Universiteit Brussel (Elsene, Belgium)

Thesis: Interoperable and Discoverable Indoor Positioning Systems

Researching hybrid- and indoor positioning systems, interoperability of location data and discoverability of services using linked data and Solid. Further research on Augmented Reality, Virtual Reality, Artificial Intelligence, deep sensor fusion and internet of things.

Master of Science (MSc), Applied Computer Science
July, 2019

Vrije Universiteit Brussel (Elsene, Belgium)

Graduated magna cum laude

Thesis: Indoor Navigation by Centralized Tracking

Parallelism, distributed systems, multicore programming, software security and indoor navigation master thesis.

Bridging Programme, Computer Science
2016 - 2018

Vrije Universiteit Brussel (Elsene, Belgium)

Bridging programme from professional bachelor (Ba) to academic bachelor (BSc).

Bachelor (Ba), Applied Computer Science
June, 2016

Erasmushogeschool Brussel (Anderlecht, Belgium)

Graduated magna cum laude

Thesis: eID IO - Applicatie Besturing

Software Security, Software Development, Agile development, business management skills. Received an award for best final work (thesis).

Software developer (student job) — *2015*

Security penetration testing of eID cloud authentication in C++.

VDW-Consulting BV. — *2013 - 2014*

Software developer (student job)

Web UI for print management software.

Volunteer Work

V.U.B. Diving Center vzw — *2025 - now*

Board Member & Secretary

Unpaid board member and secretary of the V.U.B. Diving Center vzw, responsible for administrative tasks and supporting club activities.

Key Skills

The following is a summary of key skills, including digital, social and language skills.

Language Skills

English C1 *(certificate)* **Dutch** (Native) **French** B1

Digital Skills

Research	Semantic interoperability, Semantic Web and Linked Data, Indoor robot navigation, Localisation and Positioning
Technical	Distributed systems, Software architecture, Full stack development, Software security, Software development, Database management and design
DevOps	Docker, AWS, Cybersecurity, CI/CD, Git
Programming	TypeScript, JavaScript, Java, Python, C#
Semantic Web Stack	RDF, OWL, SHACL, SPARQL, JSON-LD, Solid, Linked Data Fragments, Linked Data Notifications, SEMIC toolkit, Linked Data Event Streams, Specification design

Social Skills

Teaching & Workshops	Teaching assistant (TA), Databases, Open Information Systems, Information Visualisation, Next Generation User Interfaces, Web Technologies, Advanced Topics in Big Data, 3D Modelling and Printing
Project Management	Project lead, Agile, Scrum, Stakeholder management, Requirements engineering, End-user evaluation

Projects

The following projects are the main projects I have been part of.

ChemStream Intranet — *2014 - now*

Development of an intranet database for storing and searching chemicals and other data.

OpenHPS: An Open-Source Hybrid Positioning System

2019 - now

OpenHPS is an open source hybrid positioning system to help developers fuse various positioning technologies and algorithms. The system offers a modular data processing framework with each modules ranging from computer vision to common algorithms such as fingerprinting or data persistence of sampled data.
https://openhps.org

POSO: Positioning System Ontology

2022 - now

The positioning system ontology is a vocabulary for describing positioning systems and the techniques these systems use to determine a position. With POSO we aim to provide semantic meaning on how a positioning system is deployed and how results are calculated. https://poso.openhps.org

SemBeacon

2022 - now

SemBeacon is an ongoing project for the development of a semantic Bluetooth Low Energy beacon specification that helps to describe the beacon itself, its deployment and the positioning system used within. https://sembeacon.org

EhB Rooster

2015 - 2018

EhBRooster.be was a project aimed at improving the usability of timetables for Erasmushogeschool Brussel, gaining significant interest from students and staff. Developed in Java and PHP, the website influenced the adoption of a new timetable system before its discontinuation in 2018.
https://web.archive.org/web/20180812082356/https://ehbrooster.be/

Publications & Media Coverage

Van de Wynckel, M., & De Roock, E. (2026). OWL-SDA: A Multi-Agent System for Generating Synthetic Data to Support Ontology Evaluation. Posters, Demos, Blue Sky, and Tutorials at SEMANTiCS 2026, 4293. https://ceur-ws.org/Vol-4293/paper44.pdf
https://solid.maximvdw.be/publications/2026/paper_semantics2026_owlsda.pdf

Van de Wynckel, M., & Van Haute, G. (2026). ODDToolkit: An Open-Source Toolkit to Help Bridge the Developer Gap in Ontology Driven Design. Posters, Demos, Blue Sky, and Tutorials at SEMANTiCS 2026, 4293. https://ceur-ws.org/Vol-4293/paper46.pdf
https://solid.maximvdw.be/publications/2026/paper_semantics2026_oddtoolkit.pdf

Van Haute, G., Goedertier, S., Fannes, P., & **Van de Wynckel, M.** (2026). Chemical Identity Lost in Regulation: A Study of Semantic Interoperability in European Chemical Substance Data. Posters, Demos, Blue Sky, and Tutorials at SEMANTiCS 2026, 4293. https://ceur-ws.org/Vol-4293/paper48.pdf
https://solid.maximvdw.be/publications/2026/paper_semantics2026_chemicalsubstance.pdf

Les systèmes de positionnement intérieur interopérables permettent de localiser des personnes et des objets au sein des bâtiments — *July, 2025*
https://nesdoo.com/les-systemes-de-positionnement-interieur-interoperables-permettent-de-localiser-des-personnes-et-des-objets-au-sein-des-batiments/

Les systèmes de positionnement intérieur interopérables peuvent déterminer l'emplacement des personnes et des objets à l'intérieur des bâtiments — *July, 2025*
https://infohightech.com/les-systemes-de-positionnement-interieur-interoperables-peuvent-determiner-lemplacement-des-personnes-et-des-objets-a-linterieur-des-batiments/

VUB researcher teaches systems that determine the location of people or objects within buildings to work together — *July, 2025*

https://press.vub.ac.be/vub-researcher-teaches-systems-that-determine-the-location-of-people-or-objects-within-buildings-to-work-together

VUB-onderzoeker leert systemen die de locatie van mensen of objecten in gebouwen bepalen, samenwerken *July, 2025*
https://press.vub.ac.be/vub-onderzoeker-leert-systemen-die-de-locatie-van-mensen-of-objecten-in-gebouwen-bepalen-samen-werken

Interoperable indoor positioning systems can determine locations of people or objects inside buildings *July, 2025*
https://techxplore.com/news/2025-07-interoperable-indoor-positioning-people.html

Researcher Maxim Van De Wynckel Defends Ph.D. on Indoor Positioning Systems *July, 2025*
https://www.lifetechnology.com/blogs/life-technology-technology-news/researcher-maxim-van-de-wynckel-defends-ph-d-on-indoor-positioning-systems

Van de Wynckel, M. (2025). Interoperable and Discoverable Indoor Positioning Systems [PhD thesis]. Vrije Universiteit Brussel – WISE lab.

Van de Wynckel, M., & Signer, B. (2025). OpenHPS: A Modular Framework to Facilitate the Development of FAIR Positioning Systems. Journal of Open Source Software, 10(110), 8113. https://doi.org/10.21105/joss.08113
https://solid.maximvdw.be/publications/2025/10.21105.joss.08113.pdf

Van de Wynckel, M., & Signer, B. (2025, May). Survey on the Privacy and Transparency of Location Data. https://doi.org/10.5281/zenodo.15564050
https://zenodo.org/records/15564052

Jong Talent: Maxim Van de Wynckel (30) zoekt oplossingen om gebruikers meer controle te geven over hun locatiegegevens *March, 2025*
https://www.knack.be/nieuws/technologie/jong-talent-maxim-van-de-wynckel-30-zoekt-oplossingen-om-gebruikers-meer-controle-te-geven-over-hun-locatiegegevens/

Van de Wynckel, M., & Signer, B. (2025). Sphero Dead Reckoning and CV Tracking Dataset. Kaggle. https://doi.org/10.34740/KAGGLE/DS/6760212
https://www.kaggle.com/datasets/maximvandewynckel/sphero-dead-reckoning-and-cv-tracking-dataset

Van de Wynckel, M. (2025). Garage Positioning Dataset. Kaggle. https://doi.org/10.34740/KAGGLE/DS/6654647
https://www.kaggle.com/datasets/maximvandewynckel/garage-positioning-dataset

Malaise, Y., **Van de Wynckel, M.**, & Signer, B. (2024). Towards Distributed Intelligent Tutoring Systems Based on User-owned Progress and Performance Data. Proceedings of the 2nd Solid Symposium (SoSy'24), 3947. https://ceur-ws.org/Vol-3947/short3.pdf
https://solid.maximvdw.be/publications/towards-distributed-intelligent-tutoring-systems-based-on-user-owned-progress-and-performance-data.pdf

Van de Wynckel, M., & Signer, B. (2024). Discoverable and Interoperable Augmented Reality Environments Through Solid Pods. Proceedings of SoSy 2024 (Poster), 2nd Solid Symposium, Leuven, Belgium, May 2024, 3947. https://ceur-ws.org/Vol-3947/short2.pdf
https://solid.maximvdw.be/publications/discoverable-and-interoperable-augmented-reality-environments-through-solid-pods.pdf

Van de Wynckel, M., Valadez, I., & Signer, B. (2024). FidMark: A Fiducial Marker Ontology for Semantically Describing Visual Markers. Proceedings of The Semantic Web (ESWC 2024), 235–250. https://doi.org/10.1007/978-3-031-60635-9_14
https://solid.maximvdw.be/publications/fidmark-a-fiducial-marker-ontology-for-semantically-describing-visual-markers.pdf

Van de Wynckel, M., & Signer, B. (2023). SemBeacon: A Semantic Proximity Beacon Solution for Discovering and Detecting the Position of Physical Things. Proceedings of the 13th International Conference on the Internet of Things, IoT '23, 9–16. https://doi.org/10.1145/3627050.3627060
https://solid.maximvdw.be/publications/sembeacon-a-semantic-proximity-beacon-solution-for-discovering-and-detecting-the-position-of-physical-things.pdf

Vermunicht, B., **Van de Wynckel, M.**, & Signer, B. (2023). 802.11 Managemement frames from a public location. https://doi.org/10.5281/zenodo.8003771
https://zenodo.org/records/8003772

Hoebeke, N., **Van de Wynckel, M.**, & Signer, B. (2023). Object Tracking on a Monopoly Game Board. https://doi.org/10.5281/zenodo.7990434
https://zenodo.org/records/7990434

Van de Wynckel, M., & Signer, B. (2022). POSO: A Generic Positioning System Ontology. The Semantic Web – ISWC 2022, 231–247. https://doi.org/10.1007/978-3-031-19433-7_14
https://solid.maximvdw.be/publications/poso-a-generic-positioning-system-ontology.pdf

Van de Wynckel, M., & Signer, B. (2022). A Solid-based Architecture for Decentralised Interoperable Location Data. Proceedings of the 12th International Conference on Indoor Positioning and Indoor Navigation (IPIN'22), CEUR Workshop Proceedings, 3248. https://ceur-ws.org/Vol-3248/paper11.pdf
https://solid.maximvdw.be/publications/a-solid-based-architecture-for-decentralised-interoperable-location-data.pdf

Van de Wynckel, M., & Signer, B. (2021). Indoor Positioning Using the OpenHPS Framework. 2021 International Conference on Indoor Positioning and Indoor Navigation (IPIN), 1–8. https://doi.org/10.1109/IPIN51156.2021.9662569
https://solid.maximvdw.be/publications/indoor-positioning-using-the-openhps-framework.pdf

Van de Wynckel, M., & Signer, B. (2021, May 9). OpenHPS: Single Floor Fingerprinting and Trajectory Dataset. https://doi.org/10.5281/zenodo.4744380
https://zenodo.org/records/4744380

Van de Wynckel, M., & Signer, B. (2020). OpenHPS: An Open Source Hybrid Positioning System (WISE-2020-01). Vrije Universiteit Brussel. https://doi.org/10.48550/ARXIV.2101.05198
https://solid.maximvdw.be/publications/openhps-an-open-source-hybrid-positioning-system.pdf

Van de Wynckel, M. (2019). Indoor Navigation by Centralized Tracking [Master's thesis, Vrije Universiteit Brussel]. https://researchportal.vub.be/en/studentTheses/indoor-navigation-by-centralized-tracking
https://solid.maximvdw.be/publications/2019/Masterthesis_MaximVandeWynckel.pdf